

No. of Functional Features Included in the Solution

Date	30 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Functional Features Overview:

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Interface	Allows users to enter weather parameters for flood prediction.	Frontend	Done	10%
2	Data Preprocessing	Cleans and prepares weather data before model prediction.	Data Processing	Done	10%
3	Flood Prediction Engine	Predicts flood occurrence using trained machine learning models.	Machine Learning	Done	20%
4	XGBoost Model Integration	Integrates the best-performing XGBoost model into the application.	Machine Learning	Done	15%
5	Prediction Result Dashboard	Displays flood or no-flood prediction results.	Frontend	Done	10%
6	Flask Web Application	Provides web-based access to the prediction system.	Backend	Done	15%
7	Model Evaluation Module	Evaluates model performance and accuracy metrics.	Analytics	Done	10%
8	IBM Cloud Deployment Support	Prepares the application for scalable cloud deployment.	Deployment	In Progress	10%

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	7

Core / Must-Have Features	Flood prediction engine, Flask web application, prediction dashboard
Additional / Nice-to-Have Features	IBM Cloud deployment, future weather API integration, automated alert notifications
Features Tested & Verified	User input interface, data preprocessing, flood prediction models, XGBoost integration, prediction dashboard, Flask application, model evaluation module (7 features)

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	Prediction form and dashboard	Input forms, result pages
2	Backend / Logic	Prediction engine and preprocessing	Flask routes, ML models
3	Database / Storage	Dataset management	CSV datasets, prediction logs
4	API / Integration	Cloud integration	IBM Cloud deployment
5	Security / Authentication	Input validation	Secure handling of user inputs